



Independent Air Group, Inc.

12046 Jacaranda Ave., Suite E, Hesperia, CA 92345

Phone: (760) 949-9972 Fax: (760) 949-3972

CA License No. 845492

HVAC TEST AND BALANCE REPORT

Nanotech Energy - Fume Hood Bi-Annual

Location: 311 Otterman Drive, Chico CA 95928

Architect: Unknown

Engineer: Unknown

General Contractor: Unknown

HVAC Contractor: Owner Direct

Air Balance Technician(s) Kent Attwood

Project Number: 22-0011-FH-2

Prepared By: D. Montoya

Approved By: Richard D. Aguilera

This is to certify that **INDEPENDENT AIR GROUP, INC.** has balanced the system described in accordance with the plans, specifications, and within the limitations of the equipment and installation. The testing, balancing, and adjusting has been performed in accordance with the standard requirements and procedures of the Test, Adjusting and Balancing Bureau (TABB) and the results of these tests are herein recorded.

TABB Certified Contractor / Certified Supervisor
RICHARD D. AGUILERA

Thursday, December 8, 2022

Date

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IAG Project No.: 22-0011-FH-2

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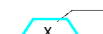
ABBREVIATIONS

ABBREVIATIONS	TERMINOLOGY
<u>A</u>	
AC:	Air Conditioner
AK:	Free Area Factor
AHU:	Air Handling Unit
AMB:	Ambient
AMP:	Amperage
ATM:	Atmosphere
AVG:	Average
<u>B</u>	
BHP:	Brake Horsepower
BTU:	British Thermal Unit
BTUH:	British Thermal Unit per Hour
<u>C</u>	
CAV:	Constant Air Volume
CD:	Ceiling Diffuser
CEG:	Ceiling Exhaust Grille
CFM:	Ceiling Exhaust Register
CH:	Cubic Feet per Minute
CHWS or CHW:	Chiller
CHWR:	Chilled Water Supply
CR:	Chilled Water Return
CRR:	Ceiling Register
Cv:	Ceiling Return Register
CV:	Specific Heat at Constant Volume
	Valve Constant
<u>D</u>	
DAT:	Discharge Air Temperature
DD:	Direct Drive
DNA:	Data Not Available
DNL:	Data Not Listed
<u>E</u>	
E:	Existing
EAT:	Entering Air Temperature
EAG:	Exhaust Air Grill
EF:	Exhaust Fan
ENT:	Entering
EG:	Exhaust Grille
ESP:	External Static Pressure

ABBREVIATIONS	TERMINOLOGY
<u>F</u>	
FCU:	Fan Coil Unit
FLA:	Full Load Amperage
FPM:	Feet per Minute
<u>G</u>	
GC:	General Contractor
GPM:	Gallons per Minute
<u>H</u>	
HC:	Heating Coil
HEPA Filter:	High Efficiency Particulate Air Filter
Hg:	Mercury
HOA:	Hand/Off/Auto Switch
HVAC:	Heating, Ventilation & Air Conditioning
<u>I</u>	
I.D.:	Inside Diameter
<u>L</u>	
LAT:	Leaving Air Temperature
LD:	Linear Diffuser
LWG:	Low Wall Grille
LWR:	Low Wall Register
LWT:	Leaving Water Temperature
<u>M</u>	
MAU or MUA:	Make-Up Air Unit or Make-Up Air
MAX:	Maximum
MB:	Mixing Box
MIN:	Minimum
<u>N</u>	
N/A:	Not applicable / No Access
NIC:	Not in Contract
NM:	Not measured
NO:	Normally Open
NTS:	Not to Scale
NVL:	No Valid Location for Testing
<u>O</u>	
O.D.:	Outside Diameter
OAT:	Outside Air Temperature
<u>P</u>	
PEF:	Power Exhaust Fan
Press.	Pressure

ABBREVIATIONS	TERMINOLOGY
<u>R</u>	
RAO:	Return Air Opening
<u>S</u>	
SA:	Stairwell Pressurization Fan
SPF:	Square
SQ:	Standard
STD:	Side Wall Grill
SWG:	Side Wall Register or Side Wall Return
SWR:	Side Wall Supply
SWS:	
<u>T</u>	
T STAT:	Thermostat
TAB:	Testing, Adjusting, and Balancing
TD:	Temperature Difference
TDH:	Total Dynamic Head
TP:	Traverse Point, Test Point or Total Pressure
TP(Motor):	Thermally Protected
TSP:	Total Static Pressure
<u>V</u>	
V:	Voltage
VAV:	Variable Air Volume
<u>W</u>	
WB:	Wet Bulb
<u>Z</u>	
Z:	Zone
Zoning:	Dividing a building into small sections for heating and cooling control, selected so that one thermostat can be requirements.

SYMBOLS

	RECTANGULAR SUPPLY AIR RISER
	RECTAGULAR SUPPLY AIR DIFFUSER
	CIRCULAR SUPPLY AIR RISER
	CIRCULAR AIR DIFFUSER
	SUPPLY FAN REFERENCE
	SUPPLY AIR DIFFUSER REFERENCE
	SUPPLY AIR VAV BOX REFERENCE
	EQUIPMENT REFERENCE
	RECTANGULAR EXHAUST AIR RISER
	RECTANGULAR EXHAUST AIR GRILLE
	CIRCULAR EXHAUST AIR RISER
	CIRCULAR EXHAUST AIR GRILLE
	EXHAUST FAN REFERENCE
	EXHAUST AIR GRILLE REFERENCE
	PITOT TUBE LOCATION
	STATIC PRESSURE SENOR
	RECTANGULAR RETURN AIR RISER
	RECTANGULAR RETURN AIR GRILLE
	CIRCULAR RETURN AIR RISER
	CICULAR RETURN AIR GRILLE
	RETURN FAN REFERENCE
	RETURN AIR GRILLE REFERENCE
	TRAVERSED FAN REFERENCE
	DUCT TRAVERSE REFERENCE
	RECTANGULAR OUTSIDE AIR RISER
	RETANGULAR OUTSIDE AIR GRILLE
	CIRCULAR OUTSIDE AIR RISER
	CIRCULAR OUTSIDE AIR GRILLE
	OUTSIDE AIR FAN REFERENCE
	OUTSIDE AIR GRILE REFERENCE

International Certification Board
Testing, Adjusting and Balancing Bureau (TABB)
Customer Satisfaction Procedure
[Independent Air Group, Inc](#)

Policy Title: Nanotech Energy - Fume Hood Bi-Annual
Date of Completion: Wednesday, November 2, 2022
Effective Until: Thursday, November 2, 2023

PURPOSE:

A formal procedure by which TAB customer complaints can be resolved promptly, fairly, and in a manner that preserves our reputation for integrity.

SCOPE:

This procedure applies to any TAB customer complaints concerning TAB work done by our company. The Customer Contact is responsible for receiving and considering all TAB customer complaints, and for implementing this procedure. Any and all customer complaints about our TAB work should be referred to the Customer Contact (identified below.) Only the Customer Contact is authorized to address TAB customer complaints.

KEY TERMS:

“Customer Contact” is the individual designated by our company to consider and respond to TAB customer complaints, and otherwise implement this procedure. The current Customer Contact is: [Richard D. Aguilera](#)

“TAB” refers to testing, adjusting and balancing work.

A “TAB customer” refers to the person (building owner or owner’s agent, general contractor, or mechanical contractor) who has retained our company to perform TAB work. However, if our company has been engaged by a contractor, we also consider the building owner of the owner’s agent as a “TAB customer” for purposes of this procedure.

NOTICE TO CUSTOMERS:

1) We will advise each person who hires our company for TAB work that we have a formal Customer Satisfaction Procedure. That notice shall be in writing. It may be by transmittal letter that returns the signed contract, or by specific reference in the contract or a contract addendum. The notice to the customer shall consist of the following (or equivalent):

Our company is dedicated to customer satisfaction. We maintain a formal Customer Satisfaction Procedure. If you have any complaint regarding our company’s testing, adjusting and balancing work, please contact Richard D. Aguilera at (760) 949-9972. That individual will make every effort to respond to your complaint by at least the end of the next business day.

2) We will respond to each TAB customer complaint as follows:

a. If the complaint can be handled by telephone consultation, we will do so. Calls will be made promptly, and courteously. There is no such thing as a “stupid question” from a customer – each and every question and complaint merits our attention and a respectful and complete answer

b. Whenever appropriate, written confirmation of advice or direction given in a telephone consultation will be promptly sent to the customer (mailed by no later than the close of the business day after consultation).

International Certification Board

Testing, Adjusting and Balancing Bureau (TABB)

Customer Satisfaction Procedure

- c.** If a customer complaint cannot be satisfactorily resolved over the telephone, a TAB technician designated by the Customer Contact will visit the customer's site at the earliest practical opportunity. Generally, a site visit will occur no later than the third business day after it is determined to be necessary. The Customer Contact may direct an earlier site visit to address urgent circumstances. In rare instances, a site visit may be postponed to a later time, if agreeable to the customer.
- d.** The technician will visit the site and assess the customer complaint(s). The technician will make every effort to be in touch with the Customer Contact (or the person designated by him or her) by telephone, from the site. If the customer's complaint can be resolved by adjustments, which the technician can perform while at the site, in a reasonable time frame, the technician will do so. If not, the technician will identify the problem(s) and make a complete report to the Customer Contact as soon as practical. Problems which cannot be resolved quickly, by the technician on site, must not be discussed with the customer at that time; rather, the technician will report those problems back to the Customer Contact for him or her to determine appropriate action.
- e.** If the visiting technician has corrected the problem, the Customer Contact will follow-up with the customer within three to five business days to ensure the customer's continuing satisfaction.
- f.** If the visiting technician has been unable to correct the problem, the Customer Contact will assess the technician's report, determine if any further testing or assessment is appropriate, and contact the customer with a complete response and/or recommendation. Normally that contact is to occur by the end of the next business day after the Customer Contact receives the visiting technician's report.
- g.** If the visiting technician's report indicates deficiencies in our TAB work, the Customer Contact will determine appropriate remedial work, determine how best to accomplish that remedial work in a prompt fashion, and will make appropriate arrangements with the customer. If our initial TAB work was faulty or inadequate, leading to the need for the remedial work, all remedial work will be without charge to the customer.
- h.** If the visiting technician's report indicates that customer action or error is the case if the customer's complaint, or that the cause is the system design, system installation, or other matter beyond our control, the Customer Contact will so advise the customer and offer reasonable assistance with appropriate follow-up. The Customer Contact will take care not to make disparaging remarks about the work of any others.
- i.** If our company does remedial TAB work, which is required because of system design, installation or other circumstance beyond our control, we will clearly inform the customer of the preparatory work that must be done, to enable us to perform high quality TAB work at a minimum additional cost to the customer.
- j.** After any remedial TAB work, whether done with or without charge, the Customer Contact shall follow-up with the customer within a reasonable period (normally three to five business days) after completion of the work, to ensure continuing customer satisfaction.
- k.** If the Customer Contact is unable to resolve a customer complaint, the Customer Contact will suggest a meeting between our [Chief Executive Officer or other appropriate officer] and a comparable representative of the customer, to attempt to resolve customer complaints short of any formal proceeding. If a resolution cannot be obtained, our company will then notify the customer in writing of the TABB Quality Assurance Program, and how to contact TABB.

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Testing, Adjusting and Balancing Bureau (TABB)

Customer Satisfaction Procedure

VARIANCES:

Although this Customer Satisfaction Procedure normally should be followed, it is not a contract. The Customer Contact is authorized to vary from this Procedure, in respect to any customer complaint. Variances should be rare exceptions, not the rule.

QUESTIONS:

Questions concerning the Customer Satisfaction Procedure, and how it is implemented, should be addressed to the Customer Contact identified above.

LIMITS:

No employee is to deal with any TAB customer complaints in a manner other than as described in this Customer Satisfaction Procedure.

CLOSING NOTES:

All employees of this company are expected at all times to be courteous to and respectful of customers. All employees are to refrain from disparaging comments about work that may have been done by others. If any employee believes that any TAB work done by our company is deficient or defective in any respect that **must not be discussed with the customer or anyone else**. Instead, report that immediately to the Customer Contact for appropriate correction consistent with this Customer Satisfaction Procedure.

International Certification Board

Testing, Adjusting and Balancing Bureau (TABB)

Customer Satisfaction Procedure

General Notes:

[Independent Air Group, Inc.](#) is a Testing and Balancing Agency that engages in testing, adjusting and balancing of air and water moving systems. [Independent Air Group, Inc.](#) does not undertake the construction of HVAC systems, fans, ductwork, terminal boxes, dampers, air distribution or other components. Additionally, our firm does not undertake construction of other building systems, such as walls, doors, windows, lights, that may have a direct affect on the performance of the system(s) being tested.

This report represents the optimal obtainable results given the limits of the installed systems and the existing conditions at the time of certification. The data and information provided herein is factual, accurate and based upon the information provided in addition to the existing conditions of the building at the time the tests were conducted.

Each system's component installation and design dictates limitations and test parameters. System specific procedures must be followed to duplicate the results listed herein. Independent Air Group, Inc. employs the standards and procedures per the latest version of the AABC National Standards and TABB Procedural Guide as applicable, unless otherwise dictated by the project specifications and/or project system specific requirements.

Verification of the report contents must be carried out within 90 days of certification and shall be accomplished by our Field Technician, using the same instruments, under the same test conditions used to develop this report; otherwise this project warranty is null and void. Our office must receive request for field verification in writing within 60 days of certification.

[Independent Air Group, Inc.](#) makes no representation or warranty that the systems and data contained herein will remain in the condition recorded beyond the date of certification. Adjustment(s), change(s) and/or tampering with component of the systems will affect the values listed herein. Changes, modifications and/or adjustment to the building structure, systems and/or sub systems, including but not limited to Architectural, Mechanical, Electrical, Plumbing, Building Automation Controls and Fire/Life Safety Alarm Systems will adversely affect the integrity of the systems and alter the data listed herein. Any such changes will void Independent Air Group, Inc.'s warranty and the customer satisfaction procedure. Proper building and HVAC systems(s) maintenance must be followed to maintain efficient conditions. Improper system maintenance directly affects system performance.

Explanation of "K" Value and Flow Hood Readings:

Air distribution was read with a Shortridge Flow Hood; Air Data Multi-Meter, model ADM-860, where possible, and where 1.0 is listed as the Effective Area, in our report, unless otherwise noted. This flow hood is a direct reading instrument that compensates and corrects for hood backpressure, altitude, density and temperature when appropriate; therefore a 1.0 is listed as the effective area. The velocity listed in when appropriate; therefore a 1.0 is listed as the effective area. The velocity listed in this report reflects the velocity through the flow hood and not the diffuser neck or face.

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Testing, Adjusting and Balancing Bureau (TABB)

Customer Satisfaction Procedure

General Procedures – All Projects:

Duct traverse readings were provided where possible and a sufficient length of straight duct was installed. Readings were not provided when one or more of the following conditions existed:

- Insufficient lengths of straight duct were installed.
- We were unable to obtain accurate readings.
- Inadequate access to the ductwork.

Scope of Work:

[Independent Air Group, Inc.](#) has been requested to provide a Survey for the above referenced project. This work is inclusive of:

1. Systems Tested and Reported in Certified report
2. Systems Tested and Mentioned in agreed Proposal
(Provided by [Independent Air Group, Inc.](#))

Project Specific Comments:

This section of our report is used to describe project special conditions, system observations and/or operational comments. At times, as applicable, items that remain outstanding and/or values that are not within the allowable project tolerances are footnoted on each individual test sheet will also be listed herein.

Independent Air Group, Inc. has noted the following upon conclusion of the Testing of this project.

[No exceptions taken.](#)

STATE OF CALIFORNIA

Contractors State License Board

Pursuant to Chapter 9 of Division 3 of the Business and Professions Code and the Rules and Regulations of the Contractors State License Board, the Registrar of Contractors does hereby issue this license to:

INDEPENDENT AIR GROUP

to engage in the business or act in the capacity of a contractor in the following classification(s):

D62 - AIR AND WATER BALANCING

Witness my hand and seal this day,
August 30, 2004

Issued August 27, 2004

Stephen P. Sands
Stephen P. Sands
Registrar of Contractors

845492

License Number

This license is the property of the Registrar of Contractors, is not transferrable, and shall be returned to the Registrar upon demand when suspended, revoked, or invalidated for any reason. It becomes void if not renewed.

Richard D. Aguilar
SIGNATURE OF LICENSEE

Richard D. Aguilar
SIGNATURE OF LICENSE QUALIFIER

International Certification Board



INTERNATIONAL
CERTIFICATION BOARD

TESTING, ADJUSTING AND BALANCING BUREAU
THE PROFESSIONAL'S CHOICE™

Sheet Metal and Air Conditioning Industry

This certifies that

Independent Air Group Inc

Has successfully completed the requirements for TABB Commissioning Contractor

CX0918011C

Certification #

Renewed: March 1, 2021

Valid Date

March 31, 2023

Expiration Date

March 1, 2021



Valid during qualified employment



[Signature]

Labor Co-Chairman

[Signature]

Management Co-Chairman

[Signature]

Administrator of ICB/TABB



#0728

ISO/IEC 17024

Personnel Certification Program

International Certification Board

Sheet Metal and Air Conditioning Industry

This certifies that

Independent Air Group Inc

Has successfully completed the requirements for TABB Contractor

January 27, 2021

BB861688C

Certification #

Renewed: January 27, 2021

Valid Date

March 31, 2023

Expiration Date



Valid during qualified employment



[Signature]

Labor Co-Chairman

[Signature]

Management Co-Chairman

[Signature]

Administrator of ICB/TABB



THE SHEET METAL INDUSTRY
INTERNATIONAL CERTIFICATION BOARD

CERTIFICATE OF RECOGNITION

AWARDED TO

Richard Aguilera

For being an Outstanding ICB-certified Professional
holding the following ICB Certification on the 19th of July, 2012 of his retirement.

TABB Supervisor and Commissioning Supervisor

Awarded the 15th of April, 2019

David Barnett

Administrator, National Energy Management Institute Committee

International Certification Board

Sheet Metal and Air Conditioning Industry

This certifies that

Kent S Attwood

Has successfully completed the requirements for TABB Supervisor

May 28, 2020

TABB1010239S

Certification #

Renewed: May 28, 2020

Valid Date

March 31, 2023

Expiration Date



Valid during qualified employment



[Signature]

Labor Co-Chairman

[Signature]

Management Co-Chairman

[Signature]

Administrator of ICB/TABB

International Certification Board

Sheet Metal and Air Conditioning Industry

This certifies that

Jesse A Gastelum

Has successfully completed the requirements for TABB Technician

March 25, 2021

BB1048598T

Certification #

Renewed: March 25, 2021

Valid Date

March 31, 2023

Expiration Date



Valid during qualified employment



[Signature]

Labor Co-Chairman

[Signature]

Management Co-Chairman

[Signature]

Administrator of ICB/TABB

Proud Member Since 11/1/2012

Member # 8220813

Membership Certificate



Shaping Tomorrow's
Built Environment Today

This is to certify

Mr Richard Don Juan Aguilera

has been granted the grade of

Associate

and is entitled to all the rights and privileges as
provided by the Constitution and Bylaws of the Society

As a member of ASHRAE Society, I comply
with the ASHRAE code of Ethics.
(www.ASHRAE.org/codeofethics).

Secretary

[Signature]
H. H. LITTLETON

President

[Signature]
THOMAS WATSON

Incorporated New York, 1895

Independent Air Group, Inc.

12046 Jacaranda Ave., Suite E, Hesperia, CA 92345

Phone: (760) 949-9972 Fax: (760) 949-3972

CA License No. 845492

HVAC Test and Balance Summary

All components of the HVAC system(s) on this project were tested, adjusted and balanced within the tolerances of + or - 10% of the design documents, with the following exceptions:

No exceptions taken.

Project: Nanotech Energy - Fume Hood Bi-Annual
 Project 22-0011-FH-2

Systems: EF-1

EF-1 (UPPER FUME HOODS & SILQ INLET)								FUME HOODS (UPPER LAB)					
SPEED	HERTZ	RPM	AMPS	VOLTS	SP (W.C.)	FMP / CFM SILQ	SILQ DAMPER	#1 FPM	#2 FPM	#3 FPM	#4 FPM	#5 FPM	#6 FPM
16%	10				0.12"	300 / 732	50%	OFF	OFF	OFF	OFF	OFF	OFF
23%	13.8	403	1.20	37.0	0.18"	290 / 707	50%	OFF	OFF	OFF	OFF	101	OFF
26%	15.2	458	1.34	47.3	0.19"	309 / 753	50%	OFF	106	OFF	OFF	95	OFF
31%	18.3	548	1.62	62.2	0.22"	417 / 1018	50%	OFF	101	OFF	101	109	OFF
38%	22.5	675	1.95	90.7	0.305"	454 / 1108	50%	OFF	110	108	OFF	102	109
41%	24.3	730	2.10	102.2	0.290"	460 / 1122	50%	110	106	101	OFF	110	100
50%	29.8	894	2.58	145.6	0.40"	562 / 1371	50%	106	99	100	99	110	101

Notes:

Tested By: Kent Attwood

Test Date: 11/2/2022

Page: 1

EF-2 (3 LARGE HOODS UPPER LAB)								FUME HOODS (UPPER LAB)					
SPEED	HERTZ	RPM	AMPS	VOLTS	SP (W.C.)			#7 FPM	#8 FPM	#9 FPM			
25%	14.6	439	1.20	45	0.259			OFF	100	OFF			
35%	20.7	621	1.84	79.8	0.42			109	104	OFF			
40%		712	2.06	98.6	0.418			100	102	106			

Notes:

Tested By: Kent Attwood Test Date: 11/2/2022

EF-4												
	HOOD #1 12 x 86	HOOD #2 12 x 86	HOOD #3 6" (.196)	HOOD #5 6" (.196)	HOOD #6 Mixer	HOOD #7 6" (.196)	HOOD #10 1.94 20 x 14 Coater	HOOD #14 62 x 12 5.17 AK Hood	HOOD #15 86 x 12 7.16AK Hood	HOOD #16 15 x 11= 1.15	ANALYTICS EF-1 24 x 24	ANALYTICS EF-2 (1) 24 x 24
TEST #1	108 FPM	110 FPM	1385 FPM 271 CFM	1277 FPM	975 FPM	1507 FPM	1162 FPM	109 EPM 564 CFM	110 FPM 787 CFM	340 FPM 391 CFM	474	574 (1)
TEST #2	107 FPM	110 FPM	1324 FPM 260 CFM	OFF	OFF	OFF	1162 CFM	107 FPM	108 FPM	365 CFM	470 CFM	582 CFM (1)
TEST #3	110	108	OFF	OFF	OFF	1552 FPM	OFF	106 FPM	110 FPM	392 CFM	482	570 (1)
TEST #4	OFF	108 FPM	OFF	OFF	OFF	OFF	OFF	108 FPM	107 FPM	OFF	480	581 (1)
TEST #5	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	475	580 (1)

Notes:

(1) Actuator set at 25% through all test

Tested By: Kent Attwood

Test Date: 11/1/2022
11/2/2022

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Project: Nanotech Energy - Fume Hood Bi-Annual
Project 22-0011-FH-2

Systems: EF-4

EF-4								
TEST	SPEED	HERTZ	RPM	AMPS	VOLTS	SP (W.C.)		
TEST #1	100%	60	1797	6.20	434.5	1.04		
TEST #2	81.5%	58.7	1761	5.70	430.0	1.01		
TEST #3	82%	49.4	1465	4.10	N/A	1.04		
TEST #4	69%	41.4	1240	3.70	N/A	1.23		
TEST #5	68%	41.97	1259	3.90	315.0	1.19		

Notes:

Independent Air Group read the fan in the worst case scenario (Test #1), and then randomly closed
or turned off actuators to verify that EF-4 maintained static pressure and the Fume Hoods maintained the required 100-110 FPM
The fan settings are as shown

Tested By: Kent Attwood

Test Date: 11/2/2022
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INDEPENDENT AIR GROUP, INC.

12046 Jacaranda Ave., Suite E
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(760) 949-9972 Fax (760) 949-3972
CA License # 845492

INSTRUMENT LIST

Date: 12/8/2022
Project Name: Nanotech Energy - Fume Hood Bi-Annual
Systems: Existing

	Instrument	Manufacturer	Model	Serial Number	Calibration Date
1	Air Data Multimeter	Shortridge	ADM 860C	M04303	2/1/2022
2	Air Data Multimeter	Shortridge	ADM-860	M89004RFS	2/4/2022
3	Air Data Multimeter	Shortridge	ADM-870	M93511	2/7/2022
4	Air Data Multimeter	Shortridge	ADM-870	M00891	2/1/2022
5	Balometer	Alnor	EBT - 731	731405001	5/12/2021
6	Digital Clamp Multimeter	Fluke	Fluke 373	50090476	1/25/2022
7	True RMS - Clamp Meter	Fluke	737	28460858	5/24/2022
8	Fluke Meter	Fluke	373	40640291	2/13/2021
9	Tachometer	Extech	RPM 33	200688680	1/25/2022
10	Digital Contact Tachometer	Jaquet	252	736549	5/25/2021
11	Hydrodata Meter	Shortridge	HDM-250	W07008	6/7/2021
12	Hydrodata Meter	Shortridge	HDM-250	W07189	6/7/2021
13	Ultrasonic Flowmeter	Regal	RH20	81103393	6/8/2021
14	Orifice Leak Detective	McGill Air	5"	486491	3/2/2022
15	Pitot Tube (*)	Dwyer	160 - 24 36 48	N/L	----
16	Rotation Vane Anemometer	Alnor	RVA801	A03381	5/21/2022
17	Sound Level Meter Type 1	Reed Instruments	IEC 651	R.118929	6/1/2022

Coded Drawing



Achieving Total Balance...
...One System At A Time

INDEPENDENT AIR GROUP INC.

12046 Jacaranda Avenue, Suite E
Hesperia, CA 92345
Office (760) 949-9972 / 949-3958
Fax (760) 949-3972

Project
Title: Nanotech Energy
311 Otterman Drive, Chico, CA

Project
Architect: Unknown

Project
Mechanical
Engineer: Unknown

HVAC
Contractor: Owner Direct

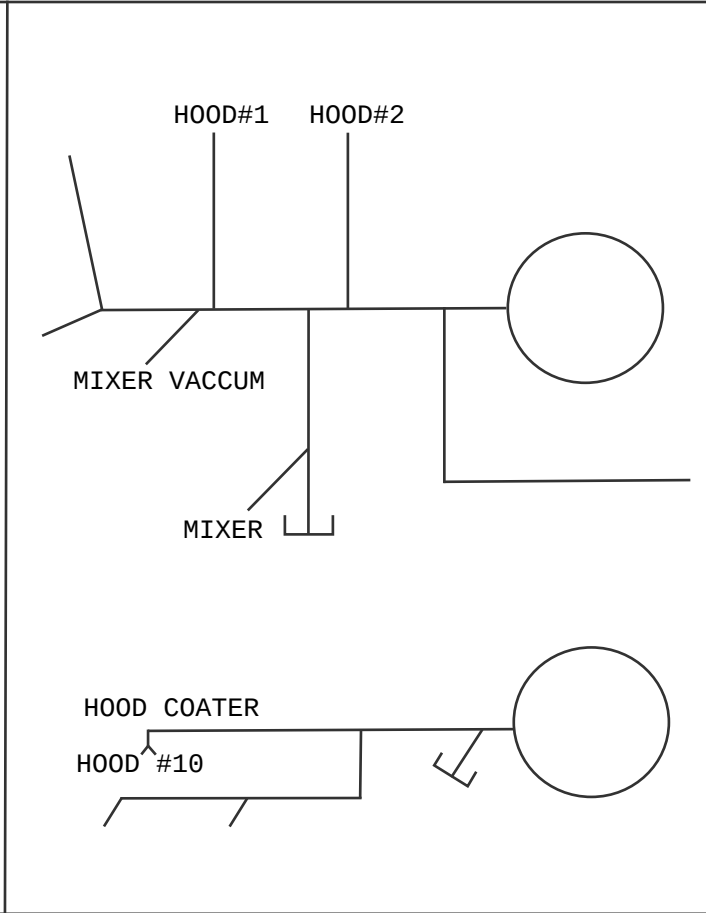
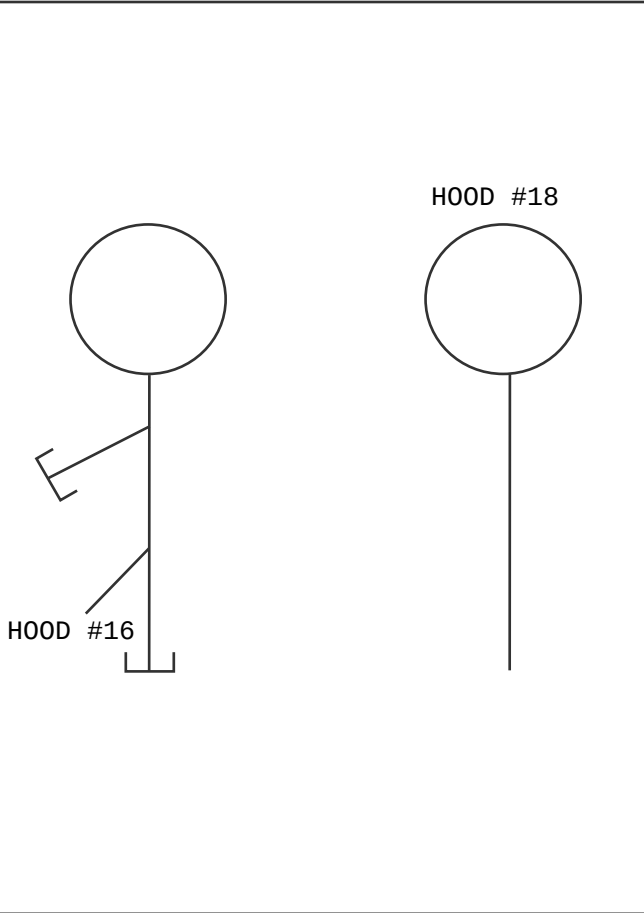
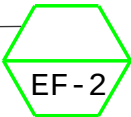
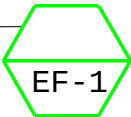
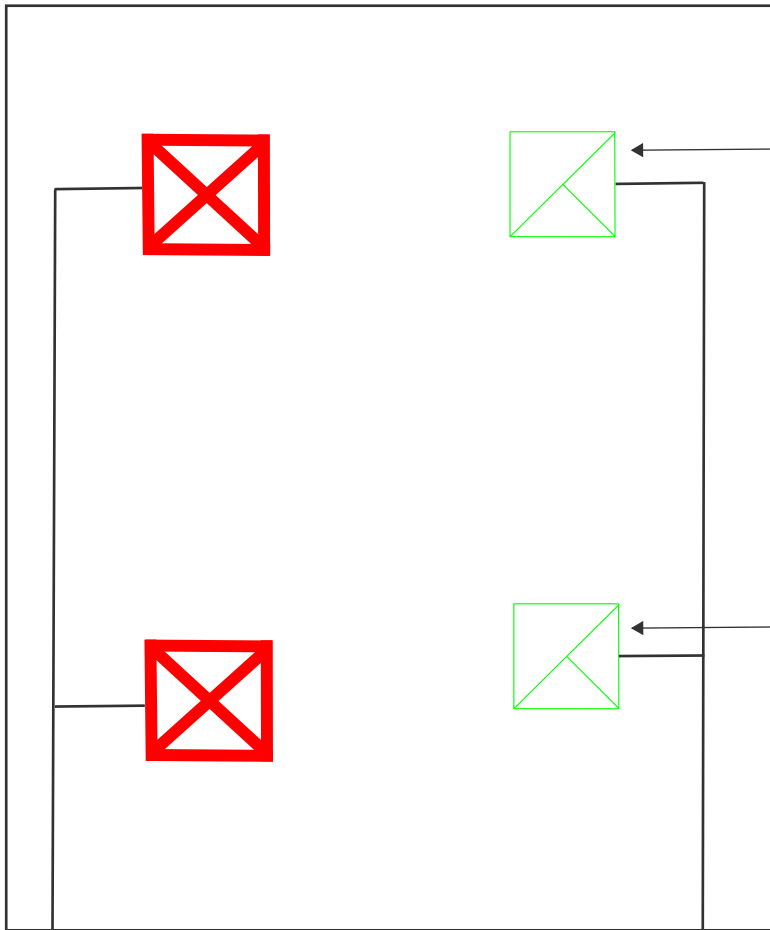
Project
Number: 22-0011-FH-2

Sheet Title: N/A

Date: 12/5/2022
Scale: N/A

Drawn By: C Aguilera
Checked By: R Aguilera

ANALYTICS



DRY LAB

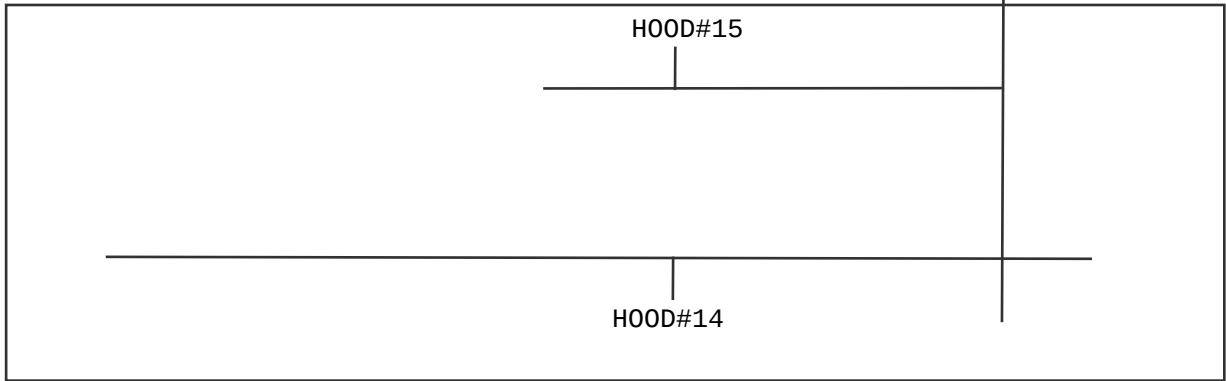
WET LAB

TO DRY LAB

TO WET LAB

EF-4

ELECTROLYTE ROOM



TEST AND BALANCE ANALYSIS LEGEND

SYMBOL	DESCRIPTIONS
	RECTAGULAR SUPPLY AIR RISER
	RECTAGULAR SUPPLY AIR DIFFUSER
	CIRCULAR SUPPLY AIR RISER
	CIRCULAR SUPPLY AIR DIFFUSER
	SUPPLY FAN REFERENCE
	SUPPLY AIR DIFFUSER REFERENCE
	SUPPLY AIR VAV BOX REFERENCE
	EQUIPMENT REFERENCE
	RECTAGULAR EXHAUST AIR RISER
	RECTAGULAR EXHAUST AIR GRILLE
	CIRCULAR EXHAUST AIR RISER
	CIRCULAR EXHAUST AIR GRILLE
	EXHAUST FAN REFERENCE
	EXHAUST AIR GRILLE REFERENCE
	PITOT TUBE LOCATION
	STATIC PRESSURE SENSOR
	RECTAGULAR RETURN AIR RISER
	RECTAGULAR RETURN AIR GRILLE
	CIRCULAR RETURN AIR RISER
	CIRCULAR RETURN AIR GRILLE
	RETURN FAN REFERENCE
	RETURN AIR GRILLE REFERENCE
	TRAVERSED FAN REFERENCE
	DUCT TRAVERSE REFERENCE
	RECTAGULAR OUTSIDE AIR RISER
	RECTAGULAR OUTSIDE AIR GRILLE
	CIRCULAR OUTSIDE AIR RISER
	CIRCULAR OUTSIDE AIR GRILLE
	OUTSIDE AIR FAN REFERENCE
	OUTSIDE AIR GRILLE REFERENCE